

Week 05 · Clear prompts and output checks

OPEN / WEEK 05

The missing invoice total

An invoice has quantity 3 and price €12, but no printed total. Should the assistant fill €36 into the extracted total field?

INVOICE / INV-04

Quantity	2
Unit price	€20
Tax	€5
Listed total	€54

Calculated total = €45. Flag the mismatch.

First vote: what would you do?

QUESTION / WEEK 05

An invoice has quantity 3 and price €12, but no printed total. Should the assistant fill €36 into the extracted total field?

Choose an initial answer and name one piece of evidence you need.

VOTE FIRST • GIVE A REASON

A Accept the first plausible result

B Check the source or measurement

Which source or test could settle this?

The journey through today's case

OPEN / WEEK 05

We will predict, connect an analogy, trace a mechanism, work an example, then explain a decision.

1

Task instruction

2

Evidence

3

Schema

4

Abstention

1. Predict • Task instruction

QUESTION / WEEK 05

What should the output contain for this invoice's listed total?

Make a choice before we explain the diagram.

VOTE FIRST • GIVE A REASON

A null, because no printed total appears.

B The assistant should infer every blank value.

Which source or test could settle this?

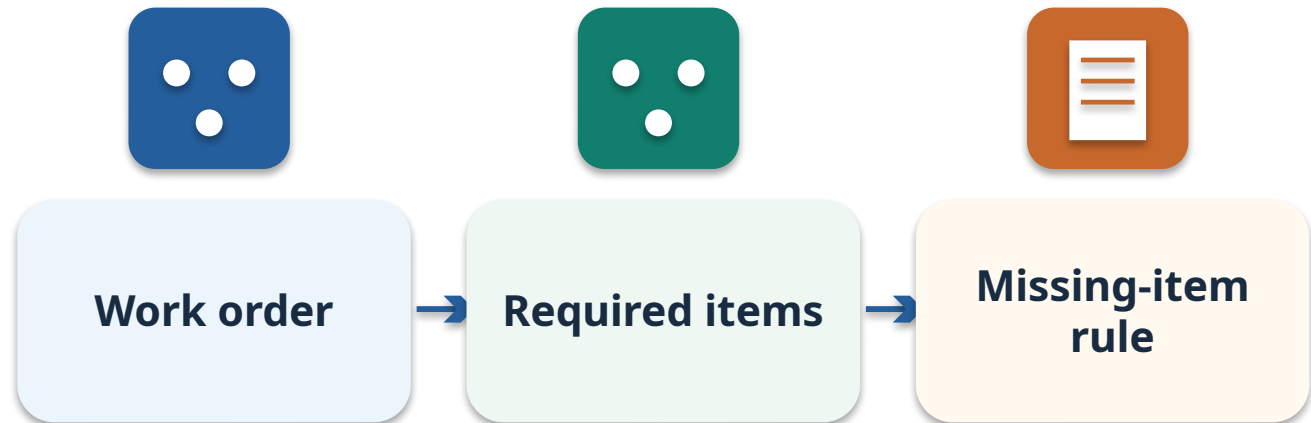
Task instruction · a familiar picture

ANALOGY / WEEK 05

A precise work order names each item and what to do if it is absent.

What matches our case, and where does the analogy stop?

A precise work order names each item and what to do if it is absent.



What matches the technical process?

Task instruction · map the parts

MAPPING / WEEK 05

Match each everyday part to the technical part.

Analogy limit: A prompt guides behaviour but does not enforce correctness by itself.

FAMILIAR SCENE

Work order

Required items

Missing-item rule

TECHNICAL IDEA

prompt

fields

null instruction

The analogy has a limit. Read it on the left.

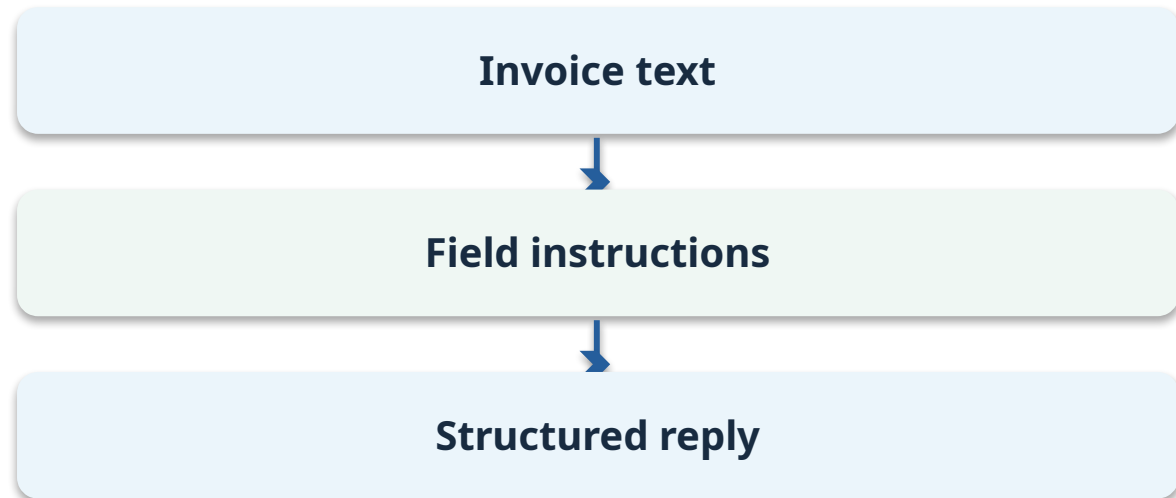
Task instruction · how it works

MECHANISM / WEEK 05

State exactly which fields to extract and how to handle missing entries.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Task instruction · work one case

WORKED / WEEK 05

Request invoice_id and listed_total; say “use null if the total is not printed.”

Point to the evidence before giving your conclusion.

GIVEN

Request invoice_id and listed_total; say “use null if the total is not printed.”

QUESTION

What should the output contain for this invoice’s listed total?

SUPPORTED ANSWER

null, because no printed total appears.

Task instruction · answer and explain

REVEAL / WEEK 05

null, because no printed total appears.

Tempting mistake: The assistant should infer every blank value.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

The assistant should infer every blank value.

SUPPORTED EXPLANATION

null, because no printed total appears.

A prompt guides behaviour but does not enforce correctness by itself.

2. Predict · Evidence

QUESTION / WEEK 05

Can €36 be labelled “extracted listed total”?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A No. It can be a separate calculated check only.

B A plausible calculation counts as document evidence.

Which source or test could settle this?

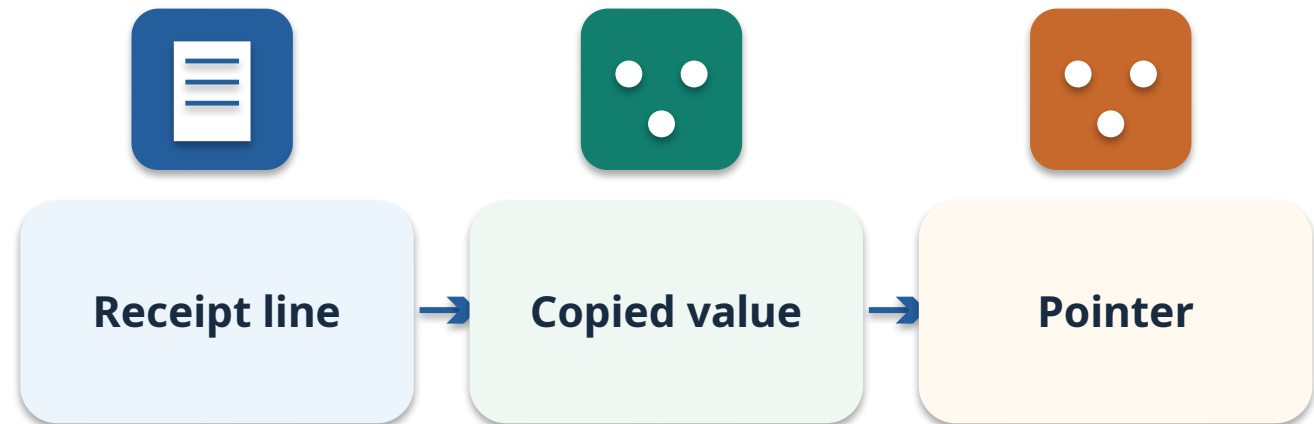
Evidence · a familiar picture

ANALOGY / WEEK 05

A clerk points to the exact line on a receipt before copying it.

What matches our case, and where does the analogy stop?

A clerk points to the exact line on a receipt before copying it.



What matches the technical process?

Evidence · map the parts

MAPPING / WEEK 05

Match each everyday part to the technical part.

Analogy limit: A source pointer may still point to the wrong line; verify the match.

FAMILIAR SCENE

Receipt line

Copied value

Pointer

TECHNICAL IDEA

source evidence

extracted field

source span

The analogy has a limit. Read it on the left.

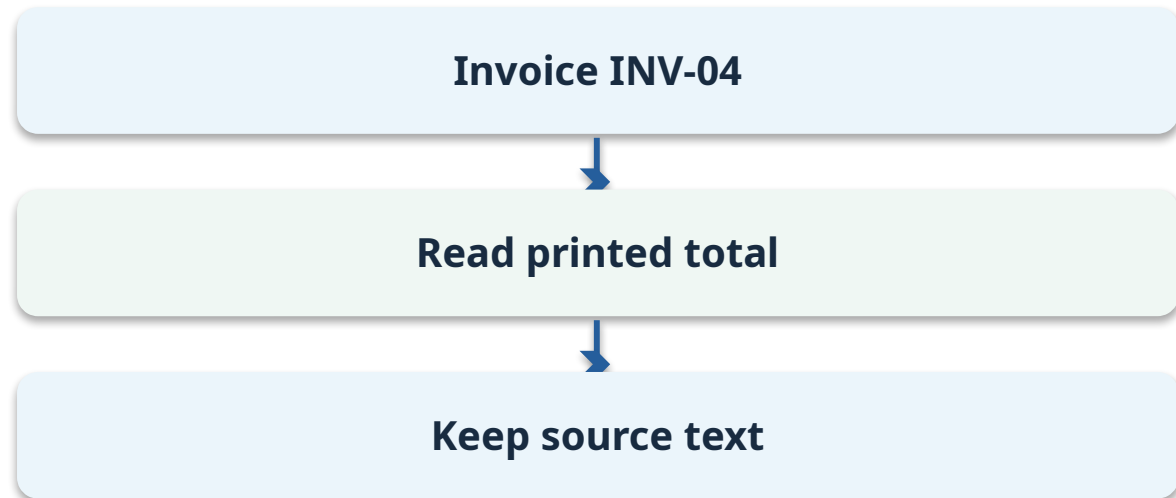
Evidence · how it works

MECHANISM / WEEK 05

Use only the supplied invoice text; do not infer a value that is absent.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Evidence · work one case

WORKED / WEEK 05

Quantity 3 and unit price €12 are visible. €36 is a calculation, not a printed total.

Point to the evidence before giving your conclusion.

GIVEN

Quantity 3 and unit price €12 are visible. €36 is a calculation, not a printed total.

QUESTION

Can €36 be labelled “extracted listed total”?

SUPPORTED ANSWER

No. It can be a separate calculated check only.

Evidence · answer and explain

REVEAL / WEEK 05

No. It can be a separate calculated check only.

Tempting mistake: A plausible calculation counts as document evidence.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

A plausible calculation counts as document evidence.

SUPPORTED EXPLANATION

No. It can be a separate calculated check only.

A source pointer may still point to the wrong line; verify the match.

3. Predict · Schema

QUESTION / WEEK 05

Which test detects a wrong but well-typed amount?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A Compare the returned amount with the original invoice text.

B Valid JSON means the facts are valid.

Which source or test could settle this?

Schema · a familiar picture

ANALOGY / WEEK 05

A shipping form checks boxes and value types before accepting a parcel.

What matches our case, and where does the analogy stop?

A shipping form checks boxes and value types before accepting a parcel.



What matches the technical process?

Schema · map the parts

MAPPING / WEEK 05

Match each everyday part to the technical part.

Analogy limit: A correct shape does not prove the value came from the document.

FAMILIAR SCENE

Boxes

Type rule

Reject stamp

TECHNICAL IDEA

field names

schema

validator flag

The analogy has a limit. Read it on the left.

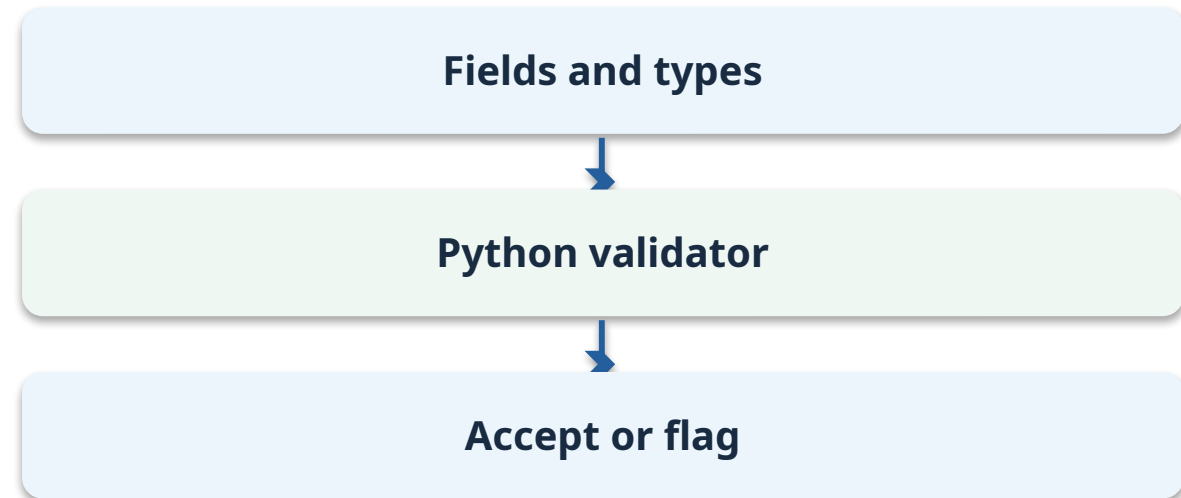
Schema · how it works

MECHANISM / WEEK 05

Validate names and types of returned fields with Python.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Schema · work one case

WORKED / WEEK 05

An ID of INV-04 and a listed total of €36 can pass type checks but fail a source check.

Point to the evidence before giving your conclusion.

GIVEN

An ID of INV-04 and a listed total of €36 can pass type checks but fail a source check.

QUESTION

Which test detects a wrong but well-typed amount?

SUPPORTED ANSWER

Compare the returned amount with the original invoice text.

Schema · answer and explain

REVEAL / WEEK 05

Compare the returned amount with the original invoice text.

Tempting mistake: Valid JSON means the facts are valid.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

Valid JSON means the facts are valid.

SUPPORTED EXPLANATION

Compare the returned amount with the original invoice text.

A correct shape does not prove the value came from the document.

4. Predict · Abstention

QUESTION / WEEK 05

Why preserve a missing value?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A It separates unknown source content from an independent calculation.

B Every missing field is a model failure to be hidden.

Which source or test could settle this?

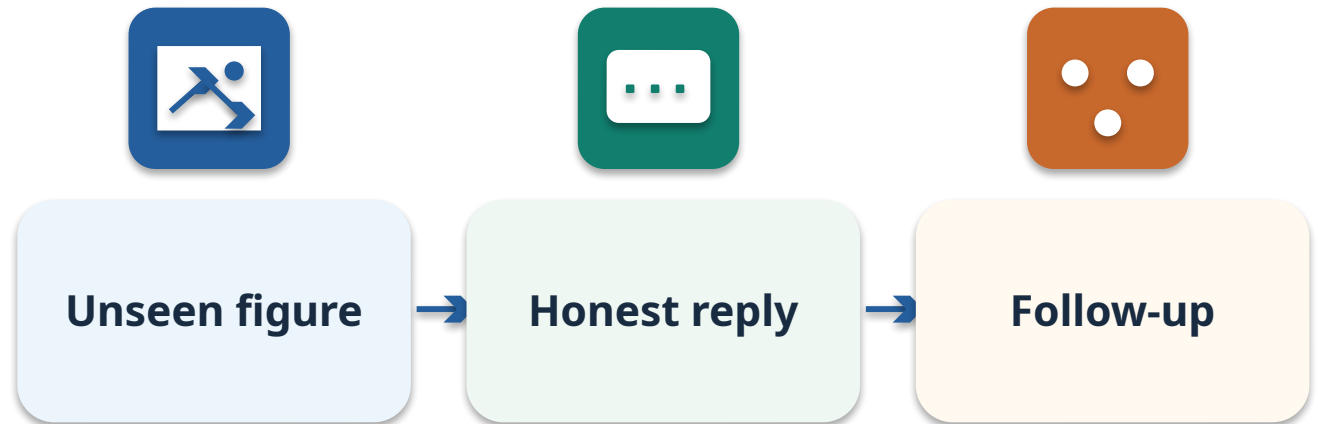
Abstention · a familiar picture

ANALOGY / WEEK 05

A careful witness says “I did not see that number.”

What matches our case, and where does the analogy stop?

A careful witness says “I did not see that number.”



What matches the technical process?

Abstention · map the parts

MAPPING / WEEK 05

Match each everyday part to the technical part.

Analogy limit: Abstention can be too cautious; measure missing and falsely missing fields.

FAMILIAR SCENE

Unseen figure

Honest reply

Follow-up

TECHNICAL IDEA

missing field

null

human review

The analogy has a limit. Read it on the left.

Abstention · how it works

MECHANISM / WEEK 05

A missing value should stay null or cause a review flag, rather than be guessed.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Abstention · work one case

WORKED / WEEK 05

When the total is absent, the validator flags null for review without inventing a figure.

Point to the evidence before giving your conclusion.

GIVEN

When the total is absent, the validator flags null for review without inventing a figure.

QUESTION

Why preserve a missing value?

SUPPORTED ANSWER

It separates unknown source content from an independent calculation.

Abstention · answer and explain

REVEAL / WEEK 05

It separates unknown source content from an independent calculation.

Tempting mistake: Every missing field is a model failure to be hidden.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

Every missing field is a model failure to be hidden.

SUPPORTED EXPLANATION

It separates unknown source content from an independent calculation.

Abstention can be too cautious; measure missing and falsely missing fields.

Evidence trail: input → result

WORKED / WEEK 05

Extract ID and listed total if present.

If total is absent, return None; 3×12 is a calculated check, not an extracted listed value.

GIVEN

Invoice INV-04 shows quantity 3 and price 12.

QUESTION

Extract ID and listed total if present.

SUPPORTED ANSWER

If total is absent, return None; 3×12 is a calculated check, not an extracted listed value.

Notebook: run the prepared example

PRACTICE / WEEK 05

Run extraction validation on ten synthetic records.

Read the original output before editing.

1

Compare a vague request with a field-specific request

2

Inspect ten synthetic invoice records

3

Validate an extraction dictionary against the schema

4

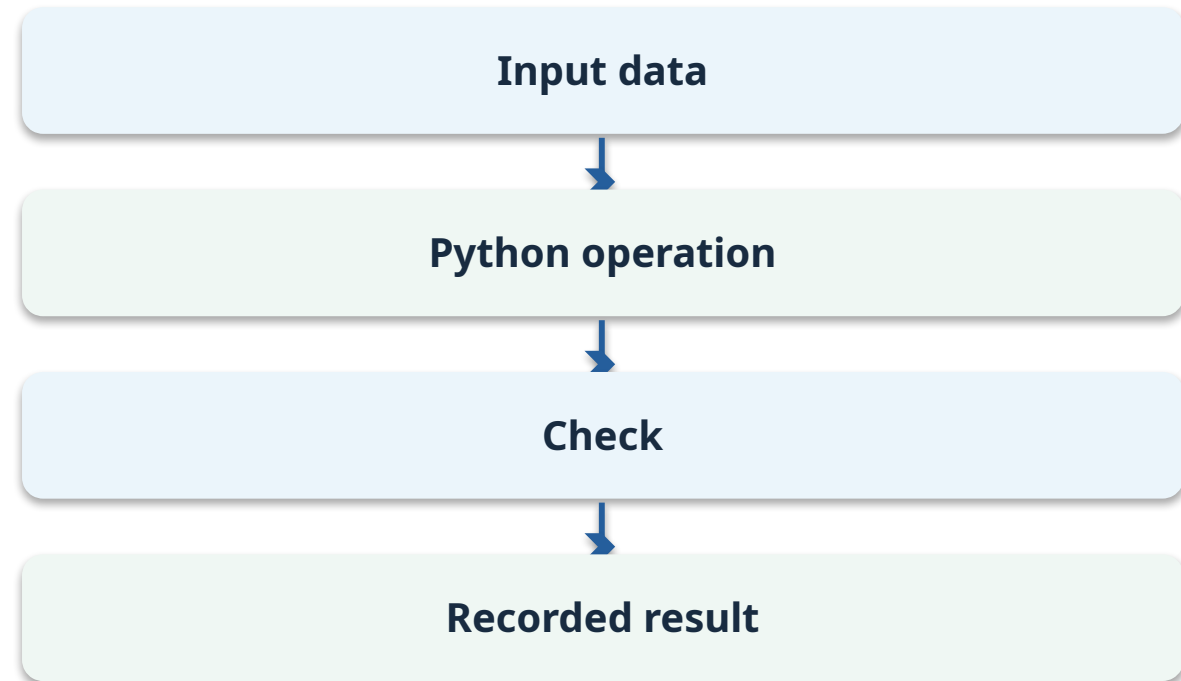
Explain a failure caused by an invented total

Find the input and check in Python

PRACTICE / WEEK 05

Locate the supplied input, the transformation, the checking line and the printed result.

FOLLOW THE DATA THROUGH EACH ARROW



Predict one controlled change

PRACTICE / WEEK 05

Delete one total and confirm the missing-field review flag fires.

What should change in the result? What stays fixed?

Before: original case

After: one change

Keep all other inputs fixed.

Run, compare and record

PRACTICE / WEEK 05

Record the input, the one changed setting, the observed result and a reason for any difference.

Input	Setting	Result
Original	baseline	record
Changed	one edit	compare

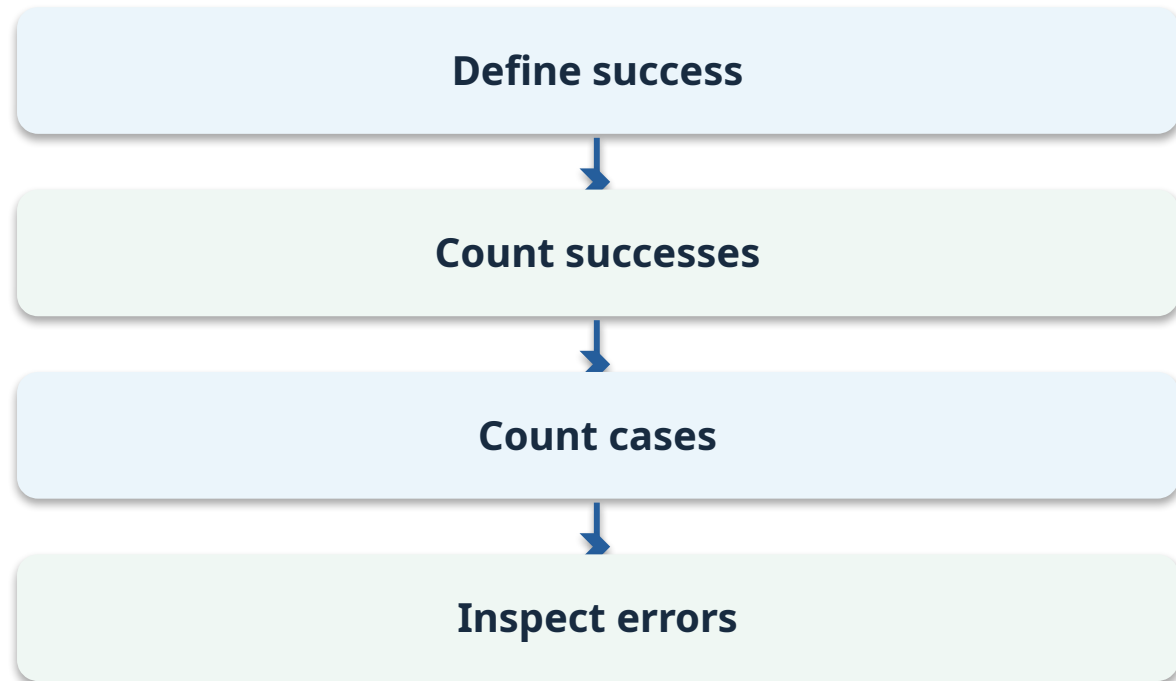
Measure without hiding the denominator

CHECK / WEEK 05

Fields correct / fields checked;
report missing fields separately

Write the numerator,
denominator and one failure
case.

FOLLOW THE DATA THROUGH EACH ARROW



Error detective: find the layer

CHECK / WEEK 05

Plausible wrong claim: Valid JSON means the invoice facts are true.

Correct it using the case evidence.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

Valid JSON means the invoice facts are true.

SUPPORTED EXPLANATION

Syntax and data types are only the first check; compare values with the original document.

Name a test that would catch this failure.

Transfer to a new situation

CHECK / WEEK 05

A syntactically valid JSON reply lists total €36. Which two checks disagree?

Explain which source or test would settle it.

VOTE FIRST • GIVE A REASON

A Use the visible evidence

B Flag missing or unsafe evidence

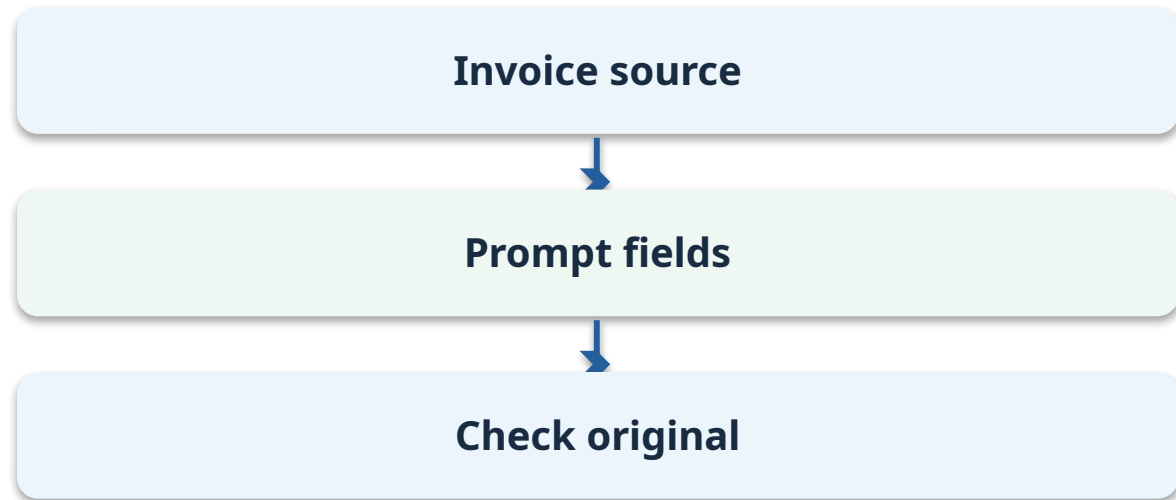
Which source or test could settle this?

Explain it to a partner

CHECK / WEEK 05

Without reading the slide, name the input, one processing step and the check. Your partner should ask “How do you know?”

FOLLOW THE DATA THROUGH EACH ARROW



Exam rehearsal · explain and apply

CHECK / WEEK 05

Write a missing-value instruction for invoice extraction.

Write a short answer, then compare with the guide.

1

State the core idea

2

Use a case detail or calculation

3

Add a limitation

4

Compare with the Word guide

Exam rehearsal · explain and apply

CHECK / WEEK 05

A JSON response contains a valid number for an absent total. Does it pass?

Write a short answer, then compare with the guide.

1

State the core idea

2

Use a case detail or calculation

3

Add a limitation

4

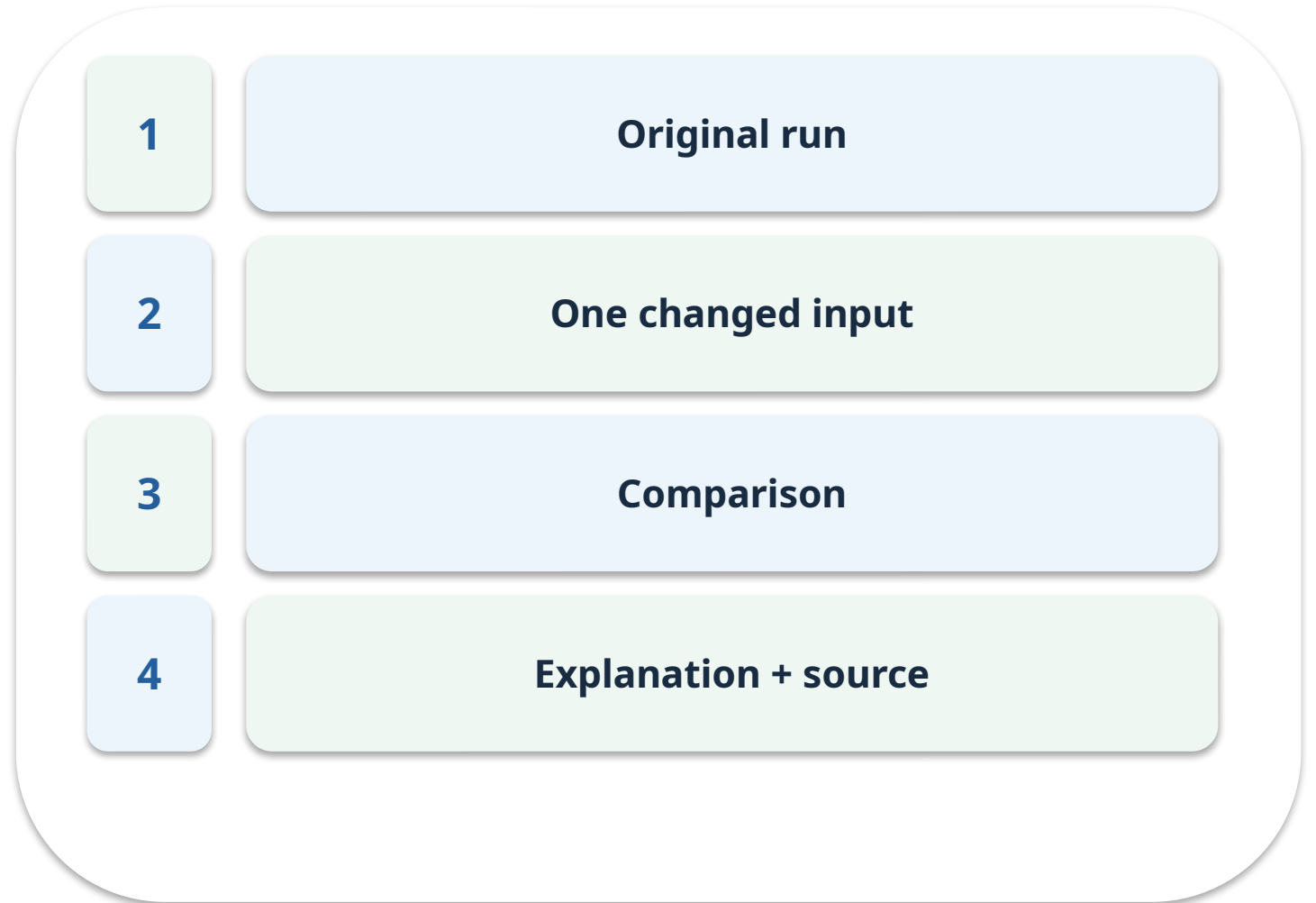
Compare with the Word guide

What to hand in

PRACTICE / WEEK 05

Prompt revision and
before/after field-check table.

Include the one changed input
and an error explanation.



Three ideas to recall next week

CHECK / WEEK 05

Specify the task, keep missing fields missing, then check structure and source.

Read the linked official sources in the Word guide and redraw one diagram from memory.

1

Task instruction

2

Evidence

3

Schema

4

Abstention